

Phase 0 pseudocode — games + algorithms (simple-modelling track)

CartPole and Flappy Bird share the same training procedures; only RESET/STEP/reward differ. REINFORCE and tabular Q-learning are implemented from scratch in the repository; A2C/PPO/DQN use the same update mathematics via Stable-Baselines3.

CartPole-v1: $s \in \mathbb{R}^4$ (pos, vel, angle, ang-vel); $A = \{0:\text{LEFT}, 1:\text{RIGHT}\}$; $r = +1/\text{step upright}$; solved ≈ 475 .

FlappyBird-v0: $s \in \mathbb{R}^{12}$; $A = \{0:\text{noop}, 1:\text{FLAP}\}$; $r = +0.1$ alive, $+1$ pipe, -1 death.

Algorithm P0.1: REINFORCE (Monte Carlo policy gradient) — coded from scratch

```
1 procedure TRAIN-REINFORCE( $env, E, \gamma, \eta$ )
2   Initialise policy parameters  $\theta$ 
3   for  $episode\ e \in \{1, \dots, E\}$  do
4      $\tau \leftarrow \text{ROLLOUT}(env, \pi_\theta)$ 
5      $\triangleright$  Discounted returns from the end of the episode
6
7      $G_T \leftarrow 0$ 
8     for  $t = T - 1, \dots, 0$  do
9        $G_t \leftarrow r_t + \gamma G_{t+1}$ 
10      Optionally normalise  $\{G_t\}$  to zero mean / unit variance
11       $L(\theta) \leftarrow -\sum_t \log \pi_\theta(a_t | s_t) G_t$ 
12       $\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$ 
13 return  $\theta$ 
```

Algorithm P0.2: Tabular Q-learning on discretized CartPole

```
1 procedure TRAIN-TABULAR-Q( $env, E, \alpha, \gamma, \epsilon$ )
2   Initialise table  $Q[\text{bin}(s), a] \leftarrow 0$ 
3    $\triangleright$  Continuous  $s$  must be binned -- does not scale to trading states
4
5   for  $episode\ e \in \{1, \dots, E\}$  do
6      $s \leftarrow env.\text{RESET}()$ 
7     while not done do
8       With prob.  $\epsilon$ :  $a \leftarrow \text{Uniform}(A)$ ; else  $a \leftarrow \arg \max_{a'} Q[\text{bin}(s), a']$ 
9        $s', r, \text{done} \leftarrow env.\text{STEP}(a)$ 
10       $y \leftarrow r + \gamma \max_{a'} Q[\text{bin}(s'), a'] \cdot (1 - \text{done})$ 
11       $Q[\text{bin}(s), a] \leftarrow Q[\text{bin}(s), a] + \alpha (y - Q[\text{bin}(s), a])$ 
12       $s \leftarrow s'$ 
13 return  $Q$ 
```

Algorithm P0.3: PPO clipped update (carried into Phase 1 trading)

```
1 procedure TRAIN-PPO( $env, steps, \gamma, \epsilon_{\text{clip}}, K$ )
2   Initialise actor  $\pi_\theta$  and critic  $V_\phi$ 
3   while  $total\ steps < budget$  do
4     Collect batch  $\mathcal{B}$  with  $\pi_\theta$ ; store also  $\log \pi_{\theta_{\text{old}}}$ 
5     Compute advantages  $A_t$ 
6     for  $epoch\ k \in \{1, \dots, K\}$  do
7        $\rho_t \leftarrow \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ 
8        $L^{\text{CLIP}}(\theta) \leftarrow -\text{mean}_t \min(\rho_t A_t, \text{clip}(\rho_t, 1 \pm \epsilon_{\text{clip}}) A_t)$ 
9        $L^V(\phi) \leftarrow \text{mean}_t (V_\phi(s_t) - G_t)^2$ 
10      Update  $\theta, \phi$  on  $L^{\text{CLIP}} + c_v L^V$ 
11       $\theta_{\text{old}} \leftarrow \theta$ 
12 return  $\theta, \phi$ 
```

DQN: same Bellman target as tabular Q with network Q_ϕ , replay buffer, and frozen target net $Q_{\bar{\phi}}$. A2C: actor loss $-\log \pi_\theta(a|s) A_t$ with $A_t = G_t - V_\phi(s)$; no clip. Suite: run Random / REINFORCE / A2C / PPO / DQN on each env_id.